Ontology-driven context engineering for semantically-aware chatbot responses

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Abstract

Large Language Models (LLMs) have achieved notable success across diverse domains, yet they remain prone to hallucinations and omissions due to their probabilistic reasoning and black-box nature. These limitations are particularly critical in high-stakes domains, where inaccurate or incomplete responses may lead to severe consequences, for example, failing to instruct a client correctly in an account activation procedure, despite having the acess to the information. Recent efforts to mitigate such issues have focused on integrating structured knowledge into LLM pipelines, predominantly through knowledge graphs. However, the selection of the relevant entities to generate a response may not be intuitive to retrieve from the graph's structure alone. The role of deeper semantic structures has not been systematically explored. Specifically, ontologies with axiomatic foundations could be used for automated inference, leading to a more precise set of entities, and therefore potentially providing a better input to generate more accurate LLM responses. This paper investigates the impact of deep semantic enrichment on LLM-based knowledge extraction and response generation. We situate our contribution within the Neuro-Symbolic AI paradigm and propose a framework that combines knowledge graphs for contextual storage with ontological reasoning grounded in structural patterns that are ontologically well-founded in UFO. By leveraging the well-founded semantic nature of ontological entities, rather than solely their graph-like structure, the proposed approach constrains interpretation and supports more precise reasoning during prompt construction. The framework is evaluated by a real company using a domain ontology of the Brazilian financial system, enriched with business rules and processes provided by an industry partner. We compare scenarios with varying strategies for leveraging semantic knowledge from the ontology, ranging from light-semantics approaches that rely only on graph-based properties, as well as a problem-based baseline designed by domain experts. Experimental results demonstrate that the Deep Semantics approach achieves superior performance in both entity extraction and response generation quality, consistently extracting a higher number of relevant entities and producing more complete and semantically aligned responses. These findings highlight the benefits of incorporating ontological depth and axiomatic semantics into LLM workflows. Overall, this work provides empirical evidence that deeper knowledge representations can significantly enhance LLM reliability and interpreting ability, advancing the development of knowledge-driven large language models.

Keywords

Ontology, Ontological Unpacking, Chatbot, Knowledge Graph, Deep Semantic Enrichment, Context Engineering

Introduction

LLMs have demonstrated remarkable ability in understanding and generating natural language, therefore often being used as chatbots, to interact and provide correct responses to users and clients, yet they remain prone to hallucinations—producing factually incorrect information. While ensuring high-quality training and reference data is vital to mitigating these errors, domain-specific data selection is equally important, as ambiguity can severely impact accuracy. This paper extends prompt engineering techniques for knowledge extraction by incorporating well-founded ontologies, enabling more effective reasoning through deeper semantic structures. By combining knowledge graphs for textual context storage with ontological unpacking for entity-level inference, our approach supports precise knowledge extraction and response formulation.

This work is situated within the broader context of Neuro-Symbolic (NeSy) AI, which integrates deep learning with symbolic reasoning. NeSy seeks to overcome the individual limitations of symbolic and sub-symbolic paradigms while leveraging their strengths—uniting two fundamental aspects of intelligent cognition: the ability to learn from experience and the capacity to reason based on learned knowledge.

Large language models (LLMs) have been successfully employed across various applications, such as natural language understanding and generation, and across a myriad of domains, such as nuclear medicine (Alberts et al. 2023), legal domain (Savelka 2023), and tourism (Carvalho and Ivanov 2024). However, these models face criticisms due

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to their reliance on memorizing information from training data, rather than accurately understanding and recalling facts put into context. This often leads to hallucinations, where the models generate outputs that contain factually incorrect or out-of-context statements (Sallam 2023; Deng and Lin 2023).

Another problem associated with LLM-generated answers is the lack of accountability, as presented by Zuccon et al. (2023). LLMs provided 86% of non-existing references, and of the remaining 14% of references that truly exist, a large part of them do not support the claims of the response. Therefore, there are various issues regarding the current performance of LLMs that stem from a lack of knowledge absorption capability, generating spurious correlations in an attempt to provide responses that emulate previously approved answers, regardless of the semantics of the generated content.

The black-box nature of LLMs also poses challenges in terms of human interpretation and explainability, as they implicitly represent knowledge within their parameters, making it difficult for human users to understand, assess, or audit the acquired knowledge. Furthermore, LLMs perform reasoning through probabilistic processes, which may yield uncertain or indecisive outcomes. Even though some LLMs employ techniques like chain-of-thought to explain their predictions Wei et al. (2022), these reasoning methods may still suffer from hallucinations, biases, and interactions with maliciously altered data, leading to unforeseen distortions in the model's response.

The limitations of LLMs become particularly concerning in high-stakes scenarios, such as medical diagnosis, or as they are widely employed as question-answering systems for customer assistance in legal and financial domains. In these contexts, an incorrect diagnosis or prediction may pose severe consequences. Explanations provided by LLMs may contradict established medical or legal knowledge. Hallucinations are of great importance in this regard and are typically discussed in the literature. However, a subtler problem may arise in the form of omissions, as was the case presented by Birkun and Gautam (2023), where the LLM did not show vital information, not capturing the complexities of the concepts involved.

McKenna et al. (2023) conducted a detailed analysis of the potential causes for hallucinations, by which they concluded that corpus-based biases are a major source for hallucinations, impacting word frequency. Corpus-based bias could be better described as the overlap between requests and the provided training data, resulting in a response with no relevant information or contradictory information, depending on the direction of the query. Therefore, the information selected to provide context is considerably relevant for LLM performance as chat bots.

Both knowledge graphs and ontologies serve as symbolic representations of knowledge (Zhang et al. 2021). Since symbolic reasoning often struggles with ambiguity and context, combining symbolic and machine learning-based approaches is a promising path forward. Symbolic AI, while precise, tends to be less efficient computationally (Shakarian et al. 2023) and less flexible with out-of-context inputs. LLMs, conversely, excel at generalization (Yang et al. 2024), but lack semantic grounding. The axiomatic nature of ontological entities can constrain user interpretation (Vasilecas 2007), and we argue that integrating this knowledge into LLMs could help limit inappropriate or inaccurate outputs.

Significant progress has been achieved in integrating structured knowledge into text generation pipelines as a potential solution to reduce hallucinations. However, most studies rely exclusively on knowledge graphs. As shown in the comparative study by Cadeddu et al. (2024), while injection techniques differ in strategy and performance, the underlying knowledge structure often remains unchanged. This reveals a gap in the literature: the impact of the depth of the knowledge structure—particularly the added axiomatic richness of ontologies—has yet to be systematically explored.

Knowledge structures have made essential contributions to the field of Natural Language Processing (NLP). Recently, integrating knowledge graphs with large language models (LLMs) has been shown to enhance performance across various tasks. However, growing attention has been directed toward more sophisticated forms of knowledge representation, such as ontologies.

Ontologies offer a semantically rich representation of expert domain knowledge, incorporating structural elements—like classes and relations—common to knowledge graphs, while adding formal axioms that define domain constraints (Guarino et al. 2009).

Despite recent progress, many approaches still overlook the value of deeper knowledge structures, and confusion between the terms **knowledge graph** and **ontology** remains prevalent in the literature—an issue that can hinder the field's development.

This work explores the state of the art, identifies emerging patterns, and clarifies distinctions between different forms of structured knowledge and their benefits for various LLM tasks. By examining these aspects, we also pinpoint the challenges and underexplored areas in advancing toward knowledge-driven large language models.

The remainder of this paper is organized in the following sections: Literature review, explaining the differences in the conceptualization of knowledge graphs and ontologies adopted and describing key foundational ontology patterns applied to the deep semantics proposal; Ontology-driven semantic enrichment, explaining the context injection framework; Experiment Design, the proposed framework is applied to various algorithms devised act as a way to grasp the effectiveness of the deep semantics proposal and the data analyzed; Results, detailing the performance of each algorithm performance in knowledge extraction and response generation quality; And a Conclusion, with the closing remarks and future work.

Literature review

This section is envisioned to standardize the conceptualization of ontologies and knowledge graphs, explain the differences between ontologies and well-founded ontlogies and detail some of the main Foundational Ontology patterns, whose proper comprehension is paramount to establishing the proposal of Deep Semantic Enrichment. Properly comprehending the differences in the knowledge structure

approaches adopted is key to understanding how "semantically deeper", that is, knowledge structures constructed based on a well-founded common universal conceptualization, can bring benefits such as more precise inference for various tasks, such as automated context injection for response-generating prompts.

Knowledge Graphs

A Knowledge Graph (KG) is a data structure that has become integral to various applications in machine learning, natural language processing, and semantic web technologies. At its core, a KG represents information in a structured format, typically using RDF, the Resource Description Framework, as defined by Ferguson and Hebels (2003), where knowledge is expressed as a set of triples; each triple consists of a head (subject), relation (predicate), and tail (object), such as (Albert Einstein, Winner Of, Nobel Prize). This structure allows for the representation of factual knowledge in a directed graph format, where nodes symbolize the entities denoting the subject and the object of the triple, and edges depict the relations between these entities. The research community often uses the terms "knowledge graph" and "knowledge base" interchangeably, depending on the context and specific characteristics emphasized in research discourse. Several scholars have attempted to outline what constitutes a KG. For example, Ehrlinger and Wöß (2016) focuses on the reasoning capabilities that KGs facilitate, highlighting their utility in AI applications that require inference and logical deduction. While Wang et al. (2017) contributes to the definition by describing a KG as a multi-relational graph that consists of entities and relations, treated as nodes and edges, respectively.

Wang et al. (2017)'s definition underscores that the relations that exist within a KG enhance its utility for complex querying and reasoning tasks. A KG can be formally defined as $G = \{E, R, F\}$, where E represents entities, R stands for relations, and F denotes the set of facts, each fact being a triple (h, r, t). One inherent characteristic of KGs that poses both challenges and opportunities for further research is their incompleteness. No knowledge base can claim to encompass all knowledge comprehensively; new entities, facts, and relationships continue to emerge, necessitating ongoing updates and revisions. Therefore, the systems must be constructed following the "Open World Assumption"(OWA). Under OWA, the knowledge base is considered inherently incomplete, meaning that the absence of information does not imply its negation (Lutz et al. 2012). Threfore systems utilizing KGs require robust mechanisms for learning, updating, and possibly inferring missing information to maintain their relevance and accuracy over time. Furthermore, the dynamic nature of knowledge creation means that KGs must be adaptable and scalable to accommodate new information efficiently and effectively.

In conclusion, while the precise definition of a knowledge graph may vary slightly among different scholars, the fundamental concept remains consistent: a graph-based representation of real-world entities and their interrelations, structured in such a way as to facilitate efficient processing, querying, and reasoning.

Ontologies

Ontologies have been used in several fields within Computer Science and correlated areas, especially due to their increasing adoption in interdisciplinary initiatives and exploring foundational abstraction levels involving cognitive psychology, philosophy of language, and linguistics (Guizzardi 2005). Therefore, it is fair to affirm that there is no consensual definition embracing all its usage contexts.

A commonly used definition in the literature and the basis for this systematic review is the one in Gruber (1993): An ontology is a formal, explicit specification of a shared conceptualization. In this work, conceptualization refers to an abstract model of the world or a specific domain, explicit implies that the elements of the ontology must be clearly defined, and formal indicates that the ontology should be machine-processable. Despite the definition not being recent, it encapsulates the essential aspects that differentiate an ontology from other knowledge structures, as will be further presented in this section.

Gruber advances the idea that an ontology is essentially the representation of the knowledge of a domain, involving a set of objects and their relations, described by a specific vocabulary. The formalization of the ontology, according to Gruber, consists of a logical theory based on an ontological commitment and a vocabulary that seeks to approximate the intended models of a domain as specified by that commitment. On the other hand, Maedche and Staab (2001) offers a complementary definition that most existing ontology representation languages are consistent with, such as Resource Description Framework (RDF) and Ontology Web Language (OWL). Maedche and Staab (2001) proposes that an ontology should be described by a five-tuple O = $\{C, R, CH, rel, OA\}$, where:

- **C** and **R** are two disjoint sets, called the set of concepts and the set of relations, respectively.
- $CH \subseteq C \times C$ is a concept hierarchy or taxonomy, where $CH(C_1, C_2)$ indicates that C_1 is a subconcept of C_2.
- $rel : R \rightarrow C \times C$ is a function that relates the concepts non-taxonomically.
- **OA** is a set of ontology axioms, expressed in an appropriate logical language.

In short, ontologies represent a fundamental tool for knowledge representation and structuring. They are structured as sets of concepts, relations, instances, and axioms, enriched by an ontological commitment, that ensures the precise and rich definitions of a domain. This structure not only represents the concepts and relations of a domain but also supports logical and deductive inference, allowing adaptability and the extraction of new knowledge in a structured manner. Thus, it is imperious to investigate the benefits that ontologies and other forms of structured knowledge may provide to Large Language Models, which, apriori, aren't capable of detecting the entities of the real world that texts refer to and are prone to hallucinate. Another rich aspect of ontologies to consider is the hierarchy of generalization, which categorizes ontologies into different levels based on scope and abstraction levels. This hierarchy consists of the upper level, task, domain, and application ontologies, each serving a different level of generalization, as depicted in

Figure1. Also, Guarino et al. (2009) proposes the following definitions for each level:

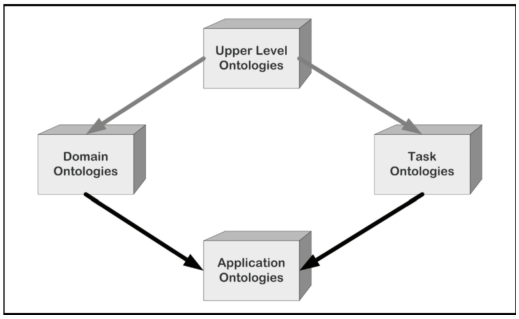


Figure 1. Ontology classification by Guarino (1998)

- *Upper-level ontologies* represent very general concepts like space, time, matter, event, etc. Which are independent of a particular domain or problem.
- *Domain ontologies* and *task ontologies* represent the vocabulary related to a generic domain or generic task or activity, by deepening and specializing the terms introduced in the upper-level ontology
- *Application ontologies* describes concepts that depend both on a domain and a task. He points out that the concepts often correspond to roles played by domain entities while a certain activity occurs.

By acknowledging these levels of generalization, ontologists can better design ontologies that are both flexible and contextually appropriate, promoting more effective data interoperability, knowledge sharing, and decision-making support across various systems and applications. This multi-tiered approach is fundamental in addressing the complexity of real-world applications, where the specificity of information varies greatly across different contexts.

The language utilized for the construction of the ontology is paramount. In this paper, we adopted OntoUML as an ontologically well-founded language for ontology-driven conceptual modeling, designed as an extension of UML (Unified Modeling Language). It incorporates ontological distinctions from a foundational ontology to ensure that conceptual models reflect real-world semantics accurately. OntoUML was developed based on the Unified Foundational Ontology (UFO), which is a foundational ontology created by Guizzardi (2005) that integrates theories from formal ontology, cognitive science, linguistics, and philosophical logic into a coherent ontological framework. By utilizing OntoUML and UFO, the system is capable of making more complex inferences, having potential gains for entity selection for LLM response generation.

Fundamental differences between ontologies and knowledge graphs

The distinction between ontologies and knowledge graphs (KGs) is essential in the domain of knowledge management and information science. While both structures aim to organize and represent knowledge, they do so with fundamentally different approaches and purposes. This subsection elucidates these differences by exploring the intrinsic characteristics of each, emphasizing their unique contributions to the field.

Ontologies are inherently philosophical, grounded in a desire to establish a universal and meticulously formal vocabulary for a particular domain. Ontologies facilitate a common understanding and enforce a rigorous structure that enhances systematic reasoning and inference capabilities. The architecture of ontologies often incorporates a hierarchical system of classification that ranges from foundational to application-specific layers. This methodological stratification allows for the delineation between general and specialized knowledge, enabling precise modeling of domains. Such a structured approach not only aids in human understanding but also enhances machine processability, thereby supporting complex logical operations and the deduction of new knowledge from facts (Maedche and Staab 2001).

Conversely, knowledge graphs prioritize practical usability and computational scalability over philosophical depth, focusing on the management and utilization of large-scale data. KGs are graph-based structures where nodes represent entities and edges depict the relations between these entities. This format is exemplified in the Resource Description Framework (RDF), which organizes knowledge in triples (head, relation, tail) that collectively form a navigable graph (Paulheim 2017).

Unlike ontologies, KGs are less concerned with strict hierarchies and are more focused on the breadth and interconnections of data. They excel in environments where rapid querying and efficient data retrieval are crucial. KGs handle a vast array of relations and entities, offering a flexible and dynamic framework that can adapt to the continuous influx of new information. This adaptability makes KGs particularly effective for applications in artificial intelligence and machine learning, where they can quickly integrate and analyze large datasets. The core distinction between ontologies and KGs lies in their targeted utility. Ontologies provide a rich, in-depth understanding of concepts, which is indispensable in scenarios requiring detailed semantic analysis and rigorous logical deductions. In contrast, KGs are better suited for handling extensive datasets where the interconnection and accessibility of information are more significant than the depth of knowledge.

Foundational Ontology patterns in OntoUML

A Foundational Ontology pattern (FOP) in OntoUML is a structure that reflects the micro-theories put forth by its underlying foundational ontology. UFO (as a top-level, foundational ontology) and OntoUML (as its isomorphich representational language) should have been defined in the subsection about ontology definitions is a system of micro-theories that addresses the main relevant classic conceptual modeling concepts, including Types and Taxonomic Structures, Attributes and Weak Entities, Datatypes, Relations, Parthood, Events, and Higher-Order types (Power Types), among others. For each of the ontological distinctions present in UFO and which are reflected as modeling constructs in OntoUML, we have a corresponding axiomatization. This axiomatization means that OntoUML constructs can only appear in a model, forming clusters of constructs with their ties and associated constraints. In other words, in OntoUML, the

modeling primitives are these pattern structures (and their corresponding axiomatization) reflecting the underlying ontological micro-theories. As a consequence, OntoUML is a pattern language whose models are constructed via the combined instantiation of the foundational patterns (Ruy et al. 2017).

Therefore, understanding some of the patterns that manifest in OntoUML is crucial to the in-depth comprehension of the proposal, as they are the basis for the rationale of the algorithm applied. These patterns appear as the entities and relations among them have their nature made explicit by the use of the OntoUML stereotypes.

The MODE pattern characterizes a MODE Universal that is linked to an ENDURANT Universal Expression through an existential dependence (inherence) relation. This ENDURANT Universal Expression serves to specify the universals whose instances act as bearers of the instances of the MODE Universal. Because a MODE may be externally dependent, the MODE pattern also includes a potentially empty set of external dependence relations that associate instances of the MODE Universal with the entities that constitute their sources of external dependence, such as linking a commitment to the entities to which that commitment refers (Ruy et al. 2017). Therefore, whenever a mode is a relevant entity, the object that it inheres in must be extracted for the proper context to be constructed.

The RELATOR pattern is defined in two variants. In the first variant, RELATORS are represented as being connected, through mediation relations—that is, existential dependence relations—to one or more textscsubstantial Universals, whose instances correspond to the entities mediated by the given RELATOR. In the second, more comprehensive variant of this pattern, a MATERIAL relation is also specified as being derived from the corresponding RELATOR Universal and established among the relata mediated by instances of that RELATOR. For example, the relation married-with can be derived from the RELATOR Universal Marriage and used to connect the ENDURANT Universals Husband and Wife, whose instances are themselves mediated by instances of Marriage. As captured by the RELATOR Pattern, the resulting MATERIAL relation may have any arity, provided it is greater than two (Ruy et al. 2017). Therefore, if a given relator is relevant to comprehend the context of a situation, all the entities mediated by it are also relevant.

The COLLECTIVE pattern characterizes a COLLECTIVE Universal together with the Universals whose instances constitute the members of these collectives. Owing to the so-called WEAK SUPPLEMENTATION PRINCIPLE, which applies to all parthood relations, the sum of the minimum cardinality constraints on the member side must be equal to or greater than two (Ruy et al. 2017). Therefore, two inferences can be made: Whenever an entity whose stereotype is a collective, it is formed by a set of members, meaning that the members are relevant as they carry the identity principle of the COLLECTIVE. Since it is known that a KIND constitutes the sole terminal symbol in the OntoUML grammar. That is, any recursive chain of definitions for these expressions in an OntoUML model terminates only upon reaching a KIND construct, therefore, to properly comprehend all the aspects of a collective, all the information that is inherited by its members up to their ultimate KIND must be collected, otherwise the collective could not have its identity principle established.

The establishment of an entity KIND as the terminal symbol, some other patterns displayed by SUBKINDS, PHASES, and ROLES stereotypes. The SUBKIND Pattern occurs in two variants. In the first, a single SUBKIND specializes a RIGID SORTAL Expression. In the second, a SUBKIND generalization set groups a disjoint (and optionally complete) collection of SUBKINDS that all specialize the same Universal, again defined by a RIGID SORTAL Expression. A SUBKIND is restricted to specializing only a RIGID SORTAL. The recursive nature of these patterns ensures that every SUBKIND, either directly or indirectly, specializes a SUBSTANCE SORTAL that supplies a uniform principle of identity for its instances (Ruy et al. 2017). Therefore, the ultimate ultimate KIND that gives the identity principle of the entity must be collected.

An ANTI-RIGID SORTAL Expression corresponds either to an instance of the PHASE Pattern or to an instance of the ROLE Pattern. The PHASE Pattern is defined by a phase partition, that is, a disjoint and complete set of two or more complementary PHASES specializing the same SORTAL, as specified by a SORTAL Expression. Once again, the recursive definition of this pattern guarantees that a SUBSTANCE SORTAL providing a common principle of identity for the instances of these PHASES is explicitly represented in the model (Ruy et al. 2017). Therefore, whenever a PHASE exists, at least one more entity with the same stereotype exists linked to the same ultimate KIND, thus all phases that fit these criteria must be collected as well as the ultimate KIND that grants the PHASES' identity principle.

Similarly, in the ROLE pattern, a ROLE specializes a SORTAL Universal, also defined through a SORTAL Expression. However, since ROLES are relationally dependent Universals, each ROLE must additionally participate in an occurrence of the RELATIONAL DEPENDENCE Pattern. The RELATIONAL DEPENDENCE Pattern is a composite pattern that characterizes the relational dependence condition of a relationally dependent Universal. Thus, a ROLE is specified by a RELATIONALLY DEPENDENT. This relational dependence is typically captured by a RELATOR pattern, where the relationally dependent Universal appears as one of the mediated types (Ruy et al. 2017). This means that whenever an entity whose stereotype is a ROLE is relevant, the RELATOR associated with it must be extracted, along with any other ROLES mediated by the same RELATORs, and for every ROLE, their ultimate KIND must be extracted due to the identity principle inherited by it.

The CATEGORY pattern can emerge in two alternative ways. In the first, a CATEGORY directly serves as a common abstraction over two or more disjoint RIGID SORTAL expressions. In the second, a CATEGORY functions indirectly as a common abstraction over another MIXIN Expression—either through a recursive occurrence of the CATEGORY pattern itself or via an instance of the ROLEMIXIN pattern—which is ultimately associated with a set of SORTAL Expressions (Ruy et al. 2017). Therefore, CATEGORIES are common abstractions of some aspect shared by disjoint objects, such as a chair and a large wooden box, being objects that a human can use to sit on; the other objects belonging to this category may be of interest for

context extraction. The ROLEMIXIN is similar to CATEGORY, the difference being that it abstracts a common aspect of two entities that are not disjoint. An example would be the ROLES of a corporate client (business) and a personal client (a person), which can be abstracted into the ROLEMIXIN of client; this implies that any properties that apply to a client also apply to a corporate client and personal client, providing a new set of entities relevant for context.

These patterns summarize the main contributions that well-founded ontologies bring to context extraction, and their application is precisely the proposal for the deep semantic enrichment for context engineering. Other relevant aspects are EVENTS, entities that typically alter the state of the world described by a SITUATION into another, meaning that both the SITUATION that triggers the event and the one that is brought about by it are relevant, along with all entities that participate in the EVENT itself. Specific association stereotypes are also very relevant, such as the historical dependence and external dependence between entities, as they are essential to construct explanations.

Ontology-Driven Semantic Enrichment

Our proposal comprises a WFO-driven semantic enrichment approach, called Deep Semantics, that uses ontologies as a leverage in LLM-based question-answering systems, to generate correct and semantically precise responses. Any form of knowledge representation that seeks to describe reality requires some level of ontological commitment, that is, an assumption about which entities exist and are relevant for the given domain being explored, how they relate, and what structural patterns emerge from the nature of the entities and relations. By having a formal WFO pattern language, relevant information can be extracted simply from the access to semantic-driven inference, in other words, because there is a well defined structure that defines the nature and relations among the stereotypes of entities, relevant complementary information can be inferred, not just from the structure, as a knowledge graph would, but actually from the foundational design patterns of the ontology, estabished by semantic formalism.

Considering that correct answers are precisely what chatbots are expected to provide when interacting with the user, the models would require an ontological level of understanding to grasp what the relevant aspects are for the principle of truth-making, defined by Guizzardi and Guarino (2024) as the foundation for explainability. A truthmaker is the relation that establishes a given fact as being true, for instance: If John is married to Jane, that means each plays the role of husband and wife, respectively; however, these roles only exist after a matrimonial relator is established. Therefore, marriage establishes their roles (Guizzardi and Guarino 2024). This means, for instance, if a system must answer who John's Wife is, it must comprehend that the relator's marriage is what establishes the counterpart role of wife to Jane; if a divorce event were to occur, this relator would be terminated, leading to a different response. Thus, explanations arise from identifying the foundational truth-making structure and entities manifested in the patterns described in a given knowledge database; this is the symbolic component of the NeSy system.

This paper introduces a framework for the semantic enrichment of chatbot responses, presenting mainly the approach of Deep Semantic enrichment, which utilizes the inherent patterns of well-founded ontologies to determine the appropriate context injection for the prompt to generate an ideal response, thus creating a NeSy system of response generation. The proposed framework comprises the following components:

1. Data gathering
2. Error classification
3. Domain expert reference response creation
4. Ontology engineering and verbalization
5. Conversation partitioning
6. Entity linking
7. Semantic enrichment
8. Prompt generation
9. Response evaluation

The goal of the Data gathering component is to collect logs of user interactions with the chatbot that resulted in a failed service experience. These failed experiences relate to any situations in which the original RAG-infused documents given to the chatbot contained the necessary information to solve the problem at hand; however, the response obtained at the time was either factually wrong or responded within the assumption of a wrong context, being factually correct but irrelevant for the problem's solution. A paramount aspect of this component is the interaction with domain experts, detailing the exact points where errors were manifested in each conversation log.

The error classification component follows immediately from the previous component; its main function is to split the errors by type. There were five potentially overlapping error categories used for this classification, based on the exact points of error, and are summarized as follows:

- Problem's nature comprehension - This was one of the most prominent types of errors, it occurred whenever, despite having enough information in the provided reference documents, the chatbot responded in another context, essentially an error derived from ambiguity, for instance, if a client informed that he is having some sort of visual glitch with the screen of his device and the chatbot responded that he should verify if the printer has any paper in it. Despite both being potential problems, they are of completely different natures.
- Hallucination - Hallucinations occur whenever the chatbot generates any response with factual errors.
- Complex errors - Complex errors are essentially the errors were originated in situations where a hypothetical correctly generated response would require a knowledge database beyond the scope of a domain-level ontology, for instance, if a customer requests an analysis regarding an accident on a given highway would impact its delivery process. This situation would require access to real-time traffic estimates, a knowledge of the company's internal delivery schedule, and initial routing. Therefore, any response classified with this type of error was deemed outside of the scope for a single domain ontology, and where not fitting as input for the subsequent components of the framework.

- Operational rules - This classification of errors is from a specific misunderstanding of internal operating rules and procedures given the particular context, this is relevant as a way to differentiate fundamentally mistaken responses, such as declaring that a driver's license is a fitting document for international travel by plane, from particularly incorrect responses, such as a flight company that requires printed boarding passes for boarding as an internal rule. These errors can only be considered when the information regarding the particular rules is available in the previous documentation; however, when generating the response, the chatbot returned something that was potentially correct, but incorrect for the context of that specific company. This distinction is important for better visualization of how to model the domain ontology that must be applied to attempt to constrain response generation.
- Mistaken response repetition - After giving a mistaken response of any of the previous error types, if the chatbot was pressed by the user with the fact that his response was ineffective or mistaken in dealing with the problem at hand, and afterwards the chatbot kept responding with the same mistaken proposed solution, this was classified as a new standalone error.

Domain experts reference response creation, which follows immediately after the error classification, and involves a series of interviews with domain experts to formulate what would be considered ideal responses. These responses will be the guidelines for the domain ontology development as well as the response evaluation. The error classification component is paramount for this response generation, as it reduces the scope, eliminating errors classified as complex or with insufficient information derived during the data gathering component. Essentially, the ontology must contain all entities described in the ideal responses, and the response quality will be made regarding completeness and correctness when compared to the reference response. The reference response is used to evaluate how good a given algorithm can navigate the knowledge structure database, extracting the relevant entities to generate the prompt, which will in turn, create the chatbot response; therefore, two types of evaluation are possible: regarding the quality of the entity extraction component and regarding the response quality itself.

The next component is ontology engineering and verbalization. It is crucial for ontology engineering that the artifact created is compliant with the syntactical rules of the foundational ontology language, thus adhering to the reference WFO; otherwise, the semantic inference would not perform appropriately. Ontology verbalization is the component of transforming the formal axioms of an ontology into natural language statements that can be easily interpreted by human readers, especially domain experts who may lack expertise in formal logic. The goal is to make the implicit meaning of logical constructs accessible and to support the validation of the modeled knowledge. In essence, ontology verbalization combines formal precision with linguistic accessibility, transforming complex logical definitions into coherent and human-readable text fragments (Ellampallil Venugopal and Kumar 2022). It is paramount that the fragments of text adequately represent all relations of a given entity, relating the stereotype of the other entities and of their relation. This way, an effective and reliable description can be created as reference input for the prompt generation, which guarantees access to the immediate context that each relevant entity carries.

Conversation partitioning is the last immediate component before enabling the start of the tests. This component consists of separating the initial interactions of the conversation, carrying the contexts up to the last moment before the final response of the chatbot, and addressing the problem. The partitioned conversations exclude the final bot's real-life response and transmit the rest of the messages to the entity linking component.

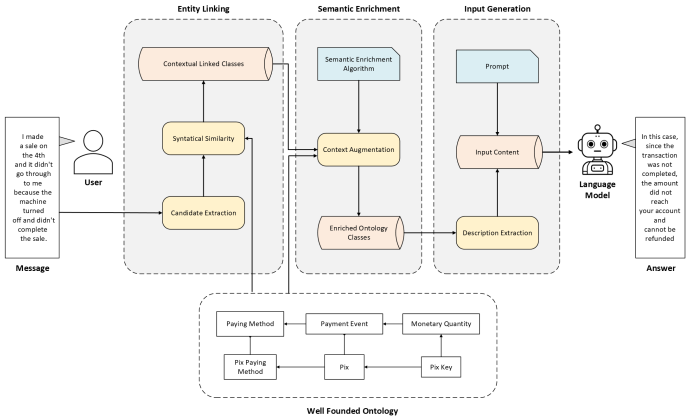


Figure 2. Semantic enrichment component

Figure 2 explains the semantic enrichment component and its surrounding components of the framework in greater detail. A message sent by the user is extracted and used as input for the entity linking component. It uses NLP techniques to extract candidate entities from the messages by standard NLP procedures, such as the removal of stop words, and all the entities are compared with the classes of the foundational ontology, selecting for the most similar classes in a syntactical threshold. The entity linking component will provide the set of Contextual Linked Classes that belong to the Well-Founded Ontology.

The Semantic enrichment is the core of the proposal. We established various algorithms to navigate the ontology and analyze the correctness of the response generation. The central hypothesis is that the algorithms will provide some improvement over the response quality as the descriptions of the classes of the ontology are collected correctly. The ontology must be well-founded so that navigation by the deep semantic algorithm can be conducted, providing context augmentation by the natures of the classes and their relations expressed in their stereotypes. In the example of Figure 2, we show an illustrative snippet of the ontology talking about Pix, the Brazilian central-bank-supported transaction method, to properly define these concepts unique to the particular context. OntoPix (Ferreira et al. 2024), a well-founded conceptual modeling in the Brazilian financial domain, was reused and further expanded upon. Therefore, based on the algorithm selected, the context augmentation component will change, creating a new set of Enriched Ontology Classes, from which the verbalisations stored as

their description will be extracted, thus defining the input content added to the prompt to produce the response.

Experiment Design

Aiming to evaluate the proposal, a series of algorithms was devised to use the structure of the knowledge graph in different patterns. These techniques, based solely on the graph structure, are the methods entitled light semantics, of which there are seven. All algorithms take as input the return of the Named Entity Recognition (NER); therefore, the entities that they consume are the same, and it is based on the previous messages of the same conversation. It is important to note that, since the experiment was conducted utilizing data from a Brazilian company, the original messages are all in Portuguese, the native language of the chatbot's clients.

Light Semantics

There are seven light algorithms:

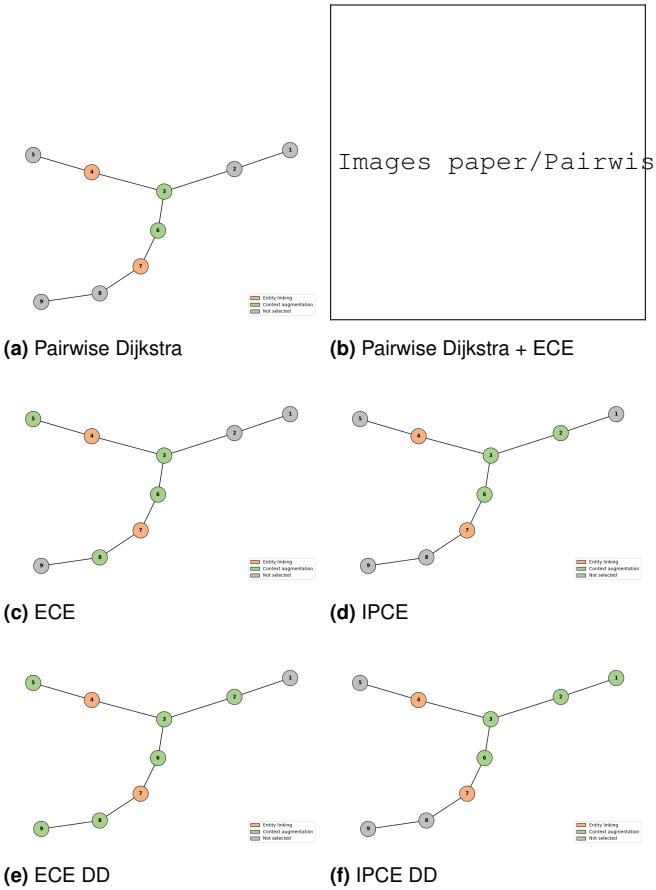
1. Pairwise Dijkstra (PDijk)
2. In-Path Context Expansion (IPCE)
3. In-Path Context Expansion Double Dive (IPCE DD)
4. Entity Context Expansion (ECE)
5. Entity Context Expansion Double Dive (ECE DD)
6. PDijk + ECE
7. Problem-Based

The PDijk follows the premise that for every pair of entities present in the Contextual Linked Classes list, the shortest path among them, collecting the entities in between, will be sufficient to provide the context augmentation. The IPCE is an expansion of the PDijk that also collects all the entities directly linked to the path formed by the pair of entities extracted by the entity linking. The entity context expansion considers all entities linked to the Contextual Linked Classes and uses them in the augmentation.

There are also the double dive variants; these algorithms will apply the same rule on the context augmented extracted entities, for instance: The IPCE Double Dive will add all entities linked to the path between the entities extracted by the entity linking. The rationale behind it is that providing more context on the connection among entities would facilitate the explanation of the entities. ECE DD collects all entities adjacent to the ones extracted by the entity linking, and then applies the same logic again, collecting the entities linked to the ones previously extracted by the ECE context augmentation. Its rationale is predicated upon gaining more context on what surrounds the entities extracted by the entity linking, therefore understanding them better.

The last graph-based algorithm is the PDijk + ECE, which applies the PDijk algorithm, collecting all entities in the shortest path, and the ECE algorithm, collecting all entities adjacent to the entities obtained from context extraction. Figure 2, shows an illustrative set of examples to better convey how the algorithms work.

The last light semantics approach is the Problem-Based algorithm. This strategy was envisioned by the pattern of typical problems that emerge naturally in the context. Domain experts were consulted and grouped subsets of the ontology as the ideal set of entities that should be contained



(a) Pairwise Dijkstra **(b)** Pairwise Dijkstra + ECE

(c) ECE **(d)** IPCE

(e) ECE DD **(f)** IPCE DD

Figure 3. Illustrative example os graph-based light semantics

in the ideal response for a given a typical recurrent set problem.

The Well-founded ontology was split into nine subsets of entities, and new classes were added since the subset is being interpreted as a typical problem, and therefore a new class, Problem, a SITUATION (Costa et al. 2006) under the Unified Foundational Ontology definition, thus describing a given possible state of the world, this situations may trigger triggers an INTERVENTION class, an EVENT as defined by Guizzardi et al. (2013), The occurrence of this intervention may bring about a new SITUATION, Solved problem. Therefore, by the inclusion of these classes and the participation of all relevant entities in the intervention event, the subsets of the ontology can be explicitly modeled in a troubleshooting framework.

The strategy of the algorithm is relatively simple; every subset of the ontology that contains all entities extracted by the entity linking is selected to be used in the prompt generation for the response. The technique proved to be effective; however, it is only functional once a typical pattern of the problem is identified and the domain experts agree with the reference solution. Therefore, it would not be applicable under the real-time need of constructing a response for a novel problem, and therefore, is not necessarily scalable. Another potential issue that may arise is the risk of extracting far more context than needed, as the identified entities at the entity linking component may be common and therefore belong to multiple unrelated problem subsets of the ontology.

This algorithm is vulnerable to hindsight bias, as defined by Kahneman (2012), which is a cognitive bias that emerges when information is selectively recalled, relaying the perception that the response was originally intuitive, when it was actually unknown at the present time. This occurs since the sets of the problem-based ontology fragments that contain the information can only be formed after various incorrect responses are experienced. This means that direct comparison to the other light semantics algorithms would be unfair, as it has the benefit of retroactive expert knowledge to split the sets of entities of interest. This approach contains a final regard that must be analyzed; it is not standardized, despite all domain exper interviewies reaching a consensus for this particular experiment, this may not always be the case, and future alterations in the ontological structure would impact the correctness of the algorithm, as new problem-based sets of ontology fragments would need to be collected. The correctness of the algorithm should be interpreted as a domain-expert-assisted entity selection system, something considerably more complex than the other light semantics methods; however, not entirely semantically aware.

Deep Semantics

The Deep Semantics approach consists of the direct application of the FOPs and DROPs to the entities found by the entity linking. If any of the entities belong to the patterns presented in Section 2, a set of semantically relevant entities can be extracted. When the descriptions of the relevant entities are injected into the prompt's context, engineering. The generated response will be semantically-aware, due to absorbing the semantically relevant entities extrapolated from the Onto-UML patterns. Therefore, the response-generating system evolves from a merely Neuronal AI system to a Neuro-Symbolic AI System.

Aiming to better illustrate how the algorithm works, consider the following real-world example: If the chatbot receives a message, "I sent the required documents; however, the account activation was not approved. Can you help me?" The entities retrieved by the entity linking were: ["Required Documents", "Account Activation"]. Figure 4 below shows the relevant ontology fragment (translated to English):

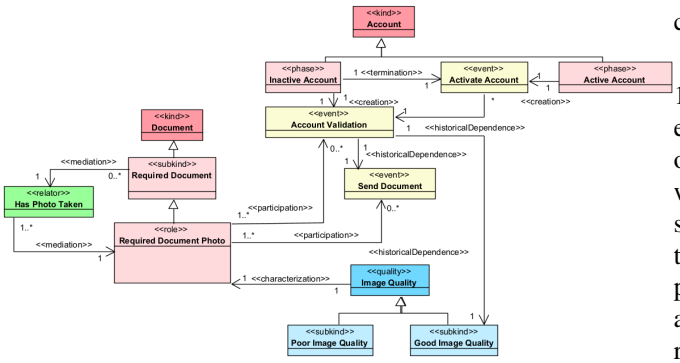


Figure 4. Ontology Fragment

Therefore, after applying the patterns, the algorithm returned the following classes for these reasons: Account Activation is an EVENT, and therefore, all entities that participate in it, are created or terminated by it, must be included, extracting ["Active Account", "Inactive Account"]. As Active Account and Inactive Account are PHASES, after applying the PHASE pattern, their ultimate KIND is extracted, obtaining the ["Account"] entity. Since the ACCOUNT ACTIVATION is an EVENT historically dependent on the Account Validation EVENT is collected.

From ACCOUNT VALIDATION, the EVENT pattern is applied again, collecting: ["Required Document Photo"], that being a ROLE has its pattern applied, collecting: ["Has Photo Taken", "Required Document", "Document"]. ACCOUNT VALIDATION is also historically dependent on ["Send Document", "Good Image Quality"]. As GOOD IMAGE QUALITY is the SUBKIND of IMAGE QUALITY, this entity is also collected, and since it is a QUALITY, the entity it characterizes must also be collected, closing the loop on REQUIRED DOCUMENT PHOTOS

With the entities collected, their descriptions are extracted from the verbalization of the ontology, allowing for the creation of the standard prompt that is structured as follows: "You are a chatbot of a credit card machine financial company, respond to the following conversation <<Message>> with the following content: <<Entities' Descriptions>>. This enables the chatbot to generate responses such as: "THE ACCOUNT ACTIVATION depends on the ACCOUNT VALIDATION, which depends on the GOOD IMAGE QUALITY of the REQUIRED DOCUMENT PHOTO. Therefore, either the photo taken does not contain the correct REQUIRED DOCUMENT or the IMAGE QUALITY of the REQUIRED DOCUMENT PHOTO is not good. Please try taking another picture or checking the one you sent."

A dataset was provided containing 60 conversations representative of the most relevant occurrences of errors when generating the responses in the past. The conversations without any error explanation were discarded, as there was not enough information to create the ontology, reducing the dataset to 45 potential conversations; of those 45, 10 of them were occurrences of complex errors demanding various domains to generate responses, and those were discarded as well. Leaving 35 potential conversations; however, some of the conversations were extremely similar, with typically two or three other instances that were virtually identical, therefore allowing the experiment to be conducted with 13 conversations, each having a six-digit identification number.

The ontology created based on these conversations had 101 classes and 258 associations, which were later shown to experts of the partner company and approved. The approved ontology was then used in a follow-up interview to define what would be the ideal responses for the 13 conversation sessions selected and the sets of entities that were expected to appear in the ideal responses. This creates a reference point that allows for the evaluation of the algorithms as a tool to collect the relevant entities and evaluate if the responses generated by their prompts are correct. After the team evaluated the responses, the team of the partner company was called again to verify whether the evaluation was valid, and they were approved unanimously. Considering the stochastic nature of LLMs to account for the risk of an unrepresentative set of responses, 30 responses were created for each conversation. The LLM utilized was the Chat GPT 4.0.

Results

The performance of the light semantics algorithms and the deep semantics algorithm must be evaluated in two dimensions: i) The quality of the algorithm as an inference machine to extract relevant entities, i.e., from the starting point that is the set of entities derived from entity linking, the algorithm must select the entities relevant to generate the response ii) The quality of the responses generated by the algorithm. Both dimensions must be evaluated separately, as the neural component of the Neuro-Symbolic AI chatbot may still be able to produce a correct response from the inferences of an incomplete set of entities. Similarly, the set of entities selected to have their descriptions injected into the prompt's context engineering may influence the likelihood of the LLM to generate hallucinations in the response.

Therefore, some metrics were derived to measure the capabilities of the algorithms in both regards. For the quality of entity extracted, the number of entities extracted was analyzed, and two metrics were derived from it: i) Completeness evaluates the entities extracted, what percentage of them belong to the reference set of entities assigned to each session by the domain experts as the appropriate entities to be used in the generation of the ideal reference response. ii) False Positive Rate was the metric derived to punish the extraction of unnecessary entities. It is calculated by dividing the number of entities extracted by the algorithm that do not belong to the ideal set of entities by the number of entities that belong to the ideal set. Therefore, the metric is highly sensitive to the number of entities of the ideal set, punishing more severely for cases few entities would be required to generate a response.

iii) correctness is the final metric applied, and it measures the percentage of correct responses generated by the prompt of each algorithm for a given conversation session. 30 responses are generated for each algorithm and session to account for the stochasticity of the LLM. The responses are considered correct if they match the ideal reference response approved by the domain experts. Table 1 shows the number of entities extracted by each algorithm.

Table 1. Number of Extracted Entities

Session/Alg	Pairwise Dijk	In-path CE	ECE	ECE DD	In-path CE DD	Dijk ECE	Problem-Based	Deep
138056	**0**	**0**	2	4	**0**	**0**	**16**	5
157753	**0**	**0**	6	17	**0**	**0**	16	**17**
138184	**0**	**0**	5	**16**	**0**	**0**	14	4
138063	**0**	**0**	5	**16**	**0**	**0**	14	7
142041	**4**	9	10	15	**35**	14	9	7
138035	**4**	6	14	**32**	9	16	24	7
138041	**4**	8	8	**22**	18	11	13	7
138053	**3**	10	10	**33**	16	17	9	4
145816	**5**	15	13	**36**	30	24	22	8
150793	**0**	**0**	3	**7**	**0**	**0**	**0**	**7**
155994	**0**	**0**	2	10	**0**	**0**	**25**	4
156042	**0**	**0**	8	32	**0**	**0**	**42**	2
Average extraction	**1.67**	4.00	7.17	20.00	9.00	6.83	17.00	6.58

Some observations can be derived from Table 1. The Dijkstra-based algorithms (Pairwise Dijkstra, In-Path CE, In-Path CE DD, and Dijk + ECE) have various instances where they returned no new entities extracted beyond the ones detected by the entity linking. This occurs whenever only one entity was extracted by the entity linking, since there is no pair of entities, no paths are produced, and no new entities can be added.

The ECE DD algorithm results in the highest average of entities extracted, followed closely by the problem-based algorithm; however, the problem-based algorithm is mostly influenced by session 156042, where only a very common entity was returned by the algorithm. This led to the collection of multiple ontology fragments and an excessive number of entities. The In-Path DD algorithm extracts an excessive number of entities frequently, whenever more than one entity is provided by the entity linking. The Deep algorithm collected a low number of entities, considering that the overall average for entity extraction is approximately 9, which is positive, as the main idea behind it was to limit excessive context injection while giving enough to respond to the questions precisely. The response quality aspect of the algorithms can be described by Tables 2 and 3.

Table 2. Ranking of the algorithms for each session

Session/Alg	Pairwise Dijk	In-path CE	ECE	ECE DD	In-path CE DD	Dijk ECE	Problem-Based	Deep
138056	5	3	6	**1**	7	4	3	2
157753	4	2	8	7	3	6	5	**1**
138184	5	5	4	3	5	5	**1**	2
138063	**1**	3	4	**1**	2	5	**1**	2
142041	4	2	3	5	5	5	5	**1**
138035	6	4	7	3	5	8	2	**1**
138041	3	2	5	4	4	5	3	**1**
138053	5	**1**	4	**1**	3	5	**1**	2
145816	3	4	4	4	4	4	2	**1**
150793	2	2	2	2	2	2	2	**1**
155994	3	3	3	3	3	3	**1**	2
156042	4	4	**1**	4	4	4	2	3
Average Rank	3.8	2.9	4.3	3.2	3.9	4.7	2.3	**1.6**
First Rank Percentage	8%	8%	8%	25%	**0%**	**0%**	33%	**50%**

From Table 2 we can see that the Deep algorithm consistently is among the three best alternatives, equating to 50% of occurrences where it was the best algorithm. As a matter of fact, the Problem-based and Deep algorithms are considerably superior, being the best performers 83% of the time, when compared to the overall performance of the graph-based light semantics algorithms.

Table 3. Percentage of correct responses for each algorithm

Session/Alg	Pairwise Dijk	In-path CE	ECE	ECE DD	In-path CE DD	Dijk ECE	Problem-Based	Deep
138056	66%	76%	**60%**	86%	**40%**	68%	76%	**86%**
157753	60%	90%	**10%**	12%	70%	50%	56%	**92%**
138184	**0%**	**0%**	23%	30%	**0%**	**0%**	**90%**	77%
138063	**0%**	80%	40%	**93%**	90%	**0%**	**93%**	90%
142041	3%	13%	26%	**0%**	**0%**	**0%**	**0%**	**100%**
138035	7%	27%	20%	40%	23%	**0%**	80%	**97%**
138041	10%	17%	36%	3%	3%	**0%**	10%	**83%**
138053	**0%**	**100%**	27%	**100%**	80%	70%	**100%**	93%
145816	3%	**0%**	50%	10%	**0%**	30%	67%	**87%**
150793	**0%**	**0%**	**0%**	**0%**	**0%**	**0%**	**0%**	**83%**
155994	**0%**	**0%**	50%	**0%**	**0%**	**0%**	**97%**	67%
156042	**0%**	**0%**	**90%**	**0%**	**0%**	50%	70%	27%
Average	20%	34%	36%	31%	26%	22%	62%	**81%**

The graph-based algorithms generate between 20% and 36% of correct responses; this seems low; however, it is important to remember that the sessions explored at the baseline, that is, with no ontological approach (pure LLM), achieved a success rate of at most 3%. The Problem-based algorithm represents a major improvement, reaching 62% correctness, while the Deep algorithm achieves a staggering 81% correctness, which is impressive, considering that the problem-based algorithm has the major advantages of domain expert intervention and hindsight bias, as discussed previously. This indicates that the Deep algorithm indeed reduced hallucination by constraining the LLM's reasoning by giving a select set of semantically relevant entities

based on their ontological nature. This perception is further detailed in Tables 4 and 5.

Table 4. Completeness percentage for each algorithm

Session/Alg	Pairwise Dijk	In-path CE	ECE	ECE DD	In-path CE DD	Dijk ECE	Deep
138056	**0%**	**0%**	18%	36%	**0%**	**0%**	**45%**
157753	**0%**	**0%**	17%	**44%**	**0%**	**0%**	33%
138184	**0%**	**0%**	33%	**88%**	**0%**	**0%**	44%
138063	**0%**	**0%**	**0%**	25%	**0%**	**0%**	**100%**
142041	**33%**	75%	42%	**100%**	**100%**	75%	66%
138035	**29%**	**29%**	71%	**93%**	**29%**	71%	50%
138041	**50%**	83%	**50%**	83%	**100%**	83%	33%
138053	**15%**	50%	30%	75%	**80%**	65%	**20%**
145816	**13%**	19%	38%	**63%**	32%	43%	50%
150793	**0%**	**0%**	**0%**	**0%**	**0%**	**0%**	**57%**
155994	**0%**	**0%**	**40%**	**40%**	**0%**	**0%**	**40%**
156042	**0%**	**0%**	20%	**70%**	**0%**	**0%**	10%
Average Completeness	**12%**	21%	30%	**60%**	28%	28%	46%
Correctness	20%	34%	36%	31%	26%	22%	**81%**

Considering the way that the Problem-based algorithm was envisioned, it would be unfair to measure its completeness and false positive rate; therefore, only the graph-based algorithms and the Deep semantic enrichment algorithm were considered.

Table 4 shows that there is no apparent correlation between completeness and correctness, as the Deep algorithm has a 46% completeness score, and nevertheless is over 2.2 times more precise than the second-best candidate. The highest average completion score belongs to the ECE DD algorithm at 60%. 100% completion occurred only 4 times, three belonging to the double dive light semantics algorithms and one to the Deep semantics algorithm.

Table 5. False positive rate for each algorithm

Session/Alg	Pairwise Dijk	In-path CE	ECE	ECE DD	In-path CE DD	Dijk ECE	Deep
138056	**0.0%**	**0%**	**0%**	**0%**	**0%**	**0%**	**0%**
157753	**0.0%**	**0%**	17%	**50%**	**0%**	**0%**	**0%**
138184	**0.0%**	**0%**	17%	**67%**	**0%**	**0%**	8%
138063	**0.0%**	**0%**	200%	**775%**	**0%**	**0%**	25%
142041	**0.0%**	**0%**	42%	**192%**	25%	42%	**0%**
138035	**0.0%**	36%	29%	**143%**	136%	64%	**0%**
138041	16.7%	50%	83%	**367%**	300%	100%	67%
138053	**0%**	**0%**	20%	**90%**	**0%**	20%	**0%**
145816	18.8%	75%	44%	**163%**	156%	106%	**0%**
150793	**0.0%**	**0%**	43%	**100%**	**0%**	**0%**	43%
155994	**0.0%**	**0%**	**0%**	**160%**	**0%**	**0%**	40%
156042	**0.0%**	**0%**	20%	**90%**	**0%**	**0%**	**0%**
Average	**3.0%**	13.4%	42.8%	**182.9%**	51.4%	27.7%	15.2%
Correctness	20%	34%	36%	31%	26%	22%	**81%**

As the false positive rate is dependent on the size of the set of entities that belong to the ideal response, there that extremely large values are reached due to a low number of entities in the ideal response, which is particularly noticeable for the ECE DD algorithm, reaching the maximum value of 775%, being the highest value for all sessions and attaining an average of 182%, over three times larger to the second largest false positive score, belonging to the other double dive algorithm. The Dijkstra-based algorithms have consistently low false positive scores, mostly due to the occurrences where no new entities are extracted. The Deep algorithm achieved a considerably low false positive score of 15.2%, placing it in the third place lowest place, between 13.4% and 27.7%. This indicates that the Deep algorithm proved to be effective in constraining the injection of undue context. Figures 5 and 6 shows the relation between completeness, false positive rate, and correctness.

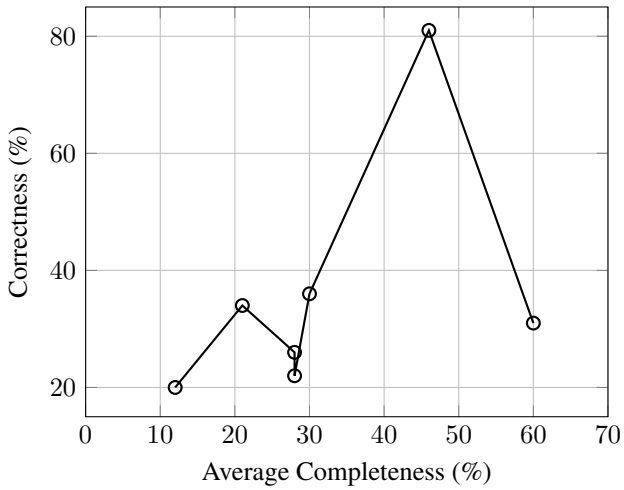


Figure 5. Correctness versus Average Completeness

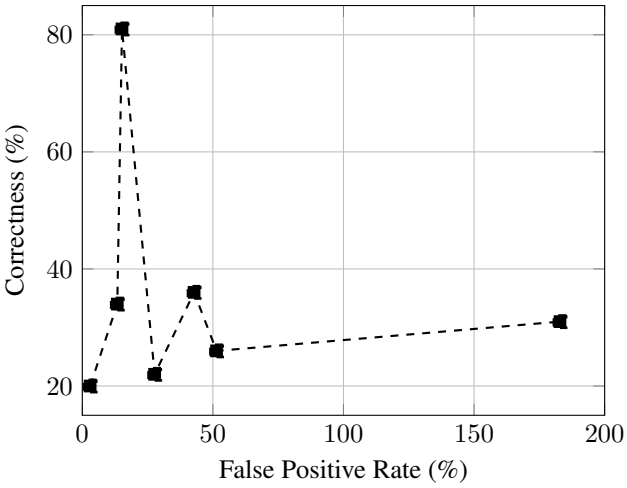


Figure 6. Correctness versus False Positive Rate

The figures clearly indicate that there is no direct correlation in both completeness and false positive rate to correctness, as the Deep algorithm reaches extremely high correctness despite only having a 46% completeness, showing that the proper selection of entities is sufficient to interact with the LLM and generate correct responses from the constrained inference component of the neural component of the system.

Simultaneously, the Deep algorithm reached a low false positive rate; however, other algorithms with lower false positive rates(FPR) are considerably worse, while the ECE DD algorithm, with the highest FPR, was not the worst performer.

There seems to be an emphasis on the quality of the entities extracted and not their quantity. Either way, the Deep semantic enrichment proved to be extremely effective, having superior performance on all relevant evaluation metrics. The gains of the deeper semantics applied can be perceived by Figures 7 and 8, comparing the average of the light semantics algorithms with the Deep correctness.

Figure 7 shows that the Deep algorithm outperforms in every metric, going from 30% to 46% completeness, reducing the false positive rate from 53.5%, and response correctness went from 28% to 81%.

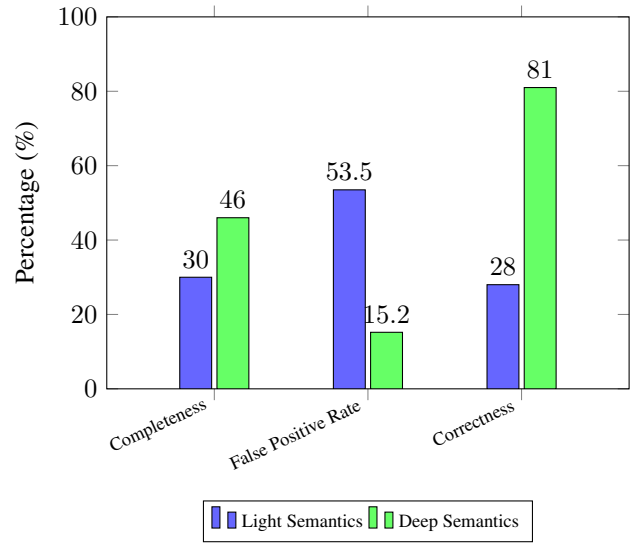


Figure 7. Comparison of completeness, false positive rate, and correctness between Light and Deep Semantics.

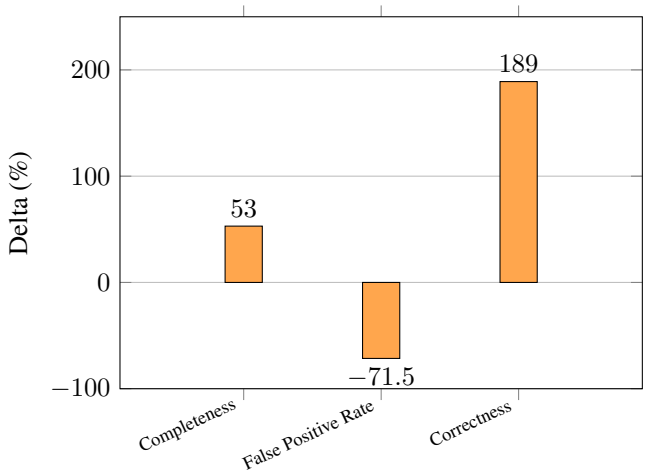


Figure 8. Relative percentage variation of Deep Semantics with respect to Light Semantics.

Figure 8 shows the percentage difference from the light semantics to the Deeep semantics metrics, having a 53% increase in completeness, a 71.5% reduction in the false positive rate, and an increase in response correctness by 189%.

Conclusion

This paper presented a framework to improve question-answering systems that combines knowledge graphs for contextual storage with ontological reasoning grounded in structural patterns that are ontologically well-founded in UFO. The proposed framework was evaluated using real conversations from a chatbot of a Brazilian company in the financial domain.

The results showed performance improvement of the Deep semantics algorithm in the perspective of entity extraction and response generation when compared to the various light semantics algorithms that make use solely of the graph-like properties of the ontology and the problem-based algorithm formulated in conjunction with the partner company's team of domain experts. Future work will evaluate further error categories by extending the domain ontology.

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